

**Editors: Dr. Azad M. Madni¹, Dr. Dinesh Verma²,
Dr. Michael Sievers¹, Ms. Marilee Wheaton¹, Dr. Edwin Ordoukhanian¹, Dr. Shatad Purohit¹**
¹University of Southern California, ²Stevens Institute of Technology

Long Beach, CA, March 18-21, 2025

A Generative AI-Driven Systems Engineering Maturity and Cost Modeling Framework

Raymond Madachy^a, Ryan Bell^b, Ryan Longshore^c

^aNaval Postgraduate School, rjmadach@nps.edu

^bNaval Postgraduate School and Naval Information Warfare Center Atlantic, ryan.bell@nps.edu

^cNaval Postgraduate School and Naval Information Warfare Center Atlantic, ryan.longshore@nps.edu

Abstract

Generative Artificial Intelligence (AI) is quickly transforming systems engineering processes in numerous ways including automating tasks, generating architecture and design options, trade-off analysis, implementation, testing, improving decision-making processes, and many more. An integrated framework is introduced with a harmonized maturity and cost model for using generative AI in systems engineering. By understanding the levels of maturity and associated cost implications of using AI, organizations can develop strategies for effectively leveraging generative AI in their engineering workflows, ensuring continuous improvement, cost reductions, and strategic integration with technological advancements. The maturity and cost model enables a cohesive evaluation of the financial impacts of AI adoption.

Keywords: *Generative AI; Large Language Models; Maturity Model; Cost Modeling; COCOMO; COSYSMO; Systems Engineering Process*

1. Introduction

A disruptive transformation is occurring on how systems engineers develop, deliver, and sustain systems due to AI assistance. Generative AI tools can support virtually all non-hardware production processes and lifecycle aspects from concept, analysis of alternatives, architecture, requirements, design, software, verification and testing, validation, etc. Drastic reductions in effort are possible using AI assistant tools that use large language models (LLMs). LLMs are a type of generative AI that utilize a deep learning algorithm responding to user queries.

However, the effective integration of generative AI into systems engineering is not a one-step process. Organizations need a structured approach to assess their readiness, plan their adoption, and track progress. To address these needs, we propose an integrated framework for process maturity and cost modeling for a comprehensive view of the technological, organizational, and financial aspects of AI adoption. The maturity model provides a roadmap for integrating generative AI, while the cost model offers a means of quantifying the associated costs and benefits for data-driven decision-making. This approach draws upon existing maturity models [1] [2] [3], and cost modeling research to update parametric cost models for AI assistance usage with empirical data [4][5].

The integration of a common 5-level maturity and cost model offers several advantages for organizations seeking to harness generative AI in their systems engineering processes. The holistic framework allows them to benchmark and plan AI adoption maturity consistently, facilitating comparisons across different

projects and industries. By aligning maturity levels with cost implications, it provides a comprehensive assessment of both the technical and financial readiness for AI adoption. Through standardized evaluation and assessment, organizations can engage in more effective strategic planning, risk management, cost estimation, and data analysis, thereby optimizing their processes and outcomes.

The 5-level model is scalable and supports organizations of various sizes in adopting AI at their own pace while progressing on a path from minimal usage to full optimization. At each level, the framework helps identify risks and cost inefficiencies, enabling teams to address challenges proactively and maximize the benefits of AI adoption. By understanding the costs and benefits of each stage, organizations can be better informed about investment decisions that align with enterprise goals. The consolidated approach also supports better prediction of systems engineering cost, schedule, and quality. It leverages parametric cost models in the public domain, ensuring transparency in the underlying formulas.

Moreover, the maturity levels and cost factor rating scale for AI assistance also serve as a foundation for collecting empirical data aligned with practices and activities specified in the ISO/IEC/IEEE15288 lifecycle standard [5]. By gathering detailed process data, organizations gain deeper insights into the effects of generative AI across a wide range of systems engineering activities and support model calibration. This iterative feedback with cost modeling refines predictive accuracy and facilitates continuous improvement, enabling organizations to optimize their AI adoption and integration strategies over time.

2. Framework Overview

The integrated framework consists of five maturity levels: Initial, Developing, Competent, Advanced, and Optimized. Each level represents a different stage of generative AI integration within systems engineering processes, characterized by increasing sophistication in AI usage, organizational preparedness, and process automation. A new parametric cost factor *AI Usage* helps assess the cost implications of AI adoption on projects. It has a 5-level rating scale describing the project behavior characteristics of generative AI practices described in the maturity levels.

Current cost modeling research [5] is calibrating the *AI Usage* cost factor to update the Constructive Systems Engineering Cost Model (COSYSMO) [7] and Constructive Cost Model (COCOMO) [8] for software development. The established COCOMO II and COSYSMO models are also precedents for having harmonized cost factor rating scales and maturity model levels.

The given names of the framework maturity levels are distinguished from those in existing maturity models, such as the Capability Maturity Model Integration (CMMI) [2] and Cybersecurity Maturity Model Certification (CMMC) 2.0 [3]. The level names are intended to better reflect the unique progression and integration challenges posed by generative AI adoption in systems engineering.

The names *Initial* and *Developing* capture the early, fragmented, and exploratory nature of integrating AI. Unlike the SEI CMMC 2.0 cybersecurity maturity model, where the primary concern is to mitigate risks as early as possible, AI integration in systems engineering focuses on discovering opportunities and incrementally building capabilities. The emphasis is on experimenting with and gaining awareness of AI's potential without a formalized strategy or extensive integration. In CMMI, the analogous stages are typically referred to as Initial (Level 1), where processes are ad hoc, and Managed (Level 2), where basic project management processes are in place, but consistent AI integration is still evolving.

The levels *Competent*, *Advanced*, and *Optimized* reflect increasing capability and strategic integration of AI into core processes. Whereas the SEI CMMC 2.0 maturity model prioritizes compliance and consistent protection, AI maturity is driven by leveraging automation, sophisticated decision-making, and continuous optimization of engineering tasks. The language emphasizes evolution, or a journey toward making AI a central component in systems engineering processes. In contrast, CMMI uses the terms Defined (Level 3), Quantitatively Managed (Level 4), and Optimizing (Level 5) to emphasize the establishment of defined processes, measurement-based management, and continuous process improvement. The names used in this AI maturity model are designed to capture the specific nature of integrating AI, focusing not only on process consistency but also on maximizing AI's role in automation, decision support, and innovation.

To progress through the maturity levels, organizations need to focus on several key enablers. These considerations include organizational infrastructure, people factors, processes and governance:

- Robust data management and access infrastructure are critical for supporting generative AI tools,

which rely on large datasets for training and inference.

- Engineers must be trained to effectively use generative AI tools. This includes not only technical skills but also an understanding of AI limitations and how to interpret AI-generated outputs.
- Successful adoption requires rethinking engineering workflows to fully integrate AI-generated insights, ensuring that AI contributes meaningfully to systems engineering goals.
- Clear policies must be established regarding the use of AI in engineering, particularly concerning issues of accountability, transparency, and bias mitigation.

3. Maturity Levels

The maturity levels outlined in this framework describe organizational processes, tools, and infrastructure. They are manifested at the project level per the cost factor rating scale in Section 4. Each maturity level corresponds to a degree of AI assistance on projects with associated cost impacts. At the project level, very large cost reductions and time savings are attained at the higher maturity levels.

Level 1: Initial

Generative AI adoption is minimal or exploratory. Organizations may experiment with AI tools in isolated projects, but there is no consistent strategy or integration into the overall engineering process. The focus at this stage is on gaining awareness of AI capabilities and assessing potential benefits. There is minimal to no AI assistance, and development relies primarily on traditional methods and tools. Consequently, the cost benefits are limited, with little to no reduction in labor or process efficiency.

Level 2: Developing

Organizations start integrating generative AI into selected engineering tasks. AI is used to assist with specific functions such as requirement analysis, preliminary design, or documentation. However, these efforts are often fragmented, with limited alignment to broader systems engineering practices. There may be moderate AI consultation for clarification or basic information retrieval. Cost implications include effort reductions for targeted tasks, but the overall financial impact is still limited due to fragmented adoption.

Level 3: Competent

Generative AI becomes a regular part of systems engineering workflows. AI tools are used across multiple phases of the lifecycle, such as conceptual design, modeling, and simulation. Organizational processes are adapted to incorporate AI outputs, and there is a growing emphasis on training engineers to work effectively with AI tools. There is regular use of AI tools for various tasks, though AI is not yet central to development. There are significant reductions in manual effort, leading to increased productivity and cost efficiency.

Level 4: Advanced

Generative AI is deeply embedded within the systems engineering processes. AI is used not only for assisting human engineers but also for autonomous generation of design alternatives, optimization, and real-time decision support. The organization has established protocols for validating AI outputs and integrating them seamlessly into engineering decision-making processes. There is frequent and strategic use of AI tools, which play a central role in multiple phases of development. Considerable cost savings are due to further reduction in manual effort, improved design quality, less rework, and faster iteration cycles.

Level 5: Optimized

Generative AI is fully integrated into the organization's systems engineering processes. AI-driven insights and automation are leveraged throughout the lifecycle, from requirements gathering to system testing and deployment. There is a strong feedback loop between AI systems and human engineers, allowing for continuous learning and improvement. At this stage, the organization may also consider ethical implications, biases, and the transparency of AI systems for responsible use of generative AI. AI tools are deeply ingrained in most development phases, crucial for decision-making, problem-solving, and automating specific tasks. At optimal efficiency, there are substantial reductions in development time and cost, making the organization highly competitive.

3.1 Maturity Level Key Practices

The integration of AI into systems engineering can be decomposed into several key practices per Table 1, each of which contributes to the overall maturity level. These practices evolve as an organization progresses through the maturity levels. They serve as critical factors that influence the success and effectiveness of AI integration at different levels. A given organization will not necessarily be at the same level for each key practice.

Practice	Level 1: Initial	Level 2: Developing	Level 3: Competent	Level 4: Advanced	Level 5: Optimized
Investment in Hardware and Infrastructure	Minimal, basic cloud subscription or server for API calls.	Low to moderate, added storage and processing for few-shot or limited RAG/fine-tuning.	Moderate, scalable storage and processing for an evolving knowledge base and RAG integration.	High, GPU-based infrastructure for extensive fine-tuning and large data processing.	Very high, robust infrastructure for multi-agent orchestration, real-time updates, and continuous learning.
Knowledge Integration and Data Management	Minimal, relying solely on model's inherent knowledge.	Basic, limited context-awareness using few-shot or small, static RAG knowledge base.	Moderate, using a fully indexed knowledge base to improve context-aware responses.	High, large, domain-specific knowledge base updated regularly for accuracy.	Real-time, cooperative agents with dynamic retrieval, high context sensitivity, and advanced data management.
LLM Customization and Fine-Tuning	None, using a generic pre-trained model without customization.	Minimal, few-shot learning or minor fine-tuning with small datasets.	Moderate, fine-tuning for specific tasks or domains with RAG integration.	High, extensive fine-tuning with proprietary or domain-specific data for customization.	Extensive, with specialized agents and continual fine-tuning for adaptability.
LLM Retraining	None, no retraining—uses only pre-trained model.	Basic periodic retraining with small data samples as needed.	Scheduled fine-tuning sessions based on new domain-specific data.	Continuous domain-specific retraining at intervals or after major data changes.	Real-time or adaptive retraining using RLHF and continuous updates with interaction.
Personnel Training	No formal training on AI tools; usage is ad hoc and based on personal initiative.	Limited training programs focused on basic AI tool usage and guidelines.	Standardized training on AI-assisted workflows, with emphasis on integrating AI into engineering tasks.	Advanced training programs incorporating best practices for AI-human collaboration and workflow optimization.	Continuous learning culture with AI literacy deeply embedded in engineering practices, with mentorship and adaptive training based on real-time AI evolution.

Table 1. Generative AI Maturity Level Practices.

Investment in Hardware and Infrastructure addresses the technological resources required for AI adoption, ranging from basic infrastructure to highly scalable, robust systems. *Knowledge Integration and Data Management* highlights how AI systems evolve from limited context awareness to fully indexed and dynamically managed knowledge bases. *LLM Customization and Fine-Tuning* describes the depth of AI model customization, from using a generic model to extensive fine-tuning with specialized datasets. *LLM Retraining* indicates the frequency and adaptiveness of model updates, evolving from minimal retraining to real-time adaptive learning with Reinforcement Learning from Human Feedback (RLHF).

Personnel Training captures the progressive development of workforce competencies to improve AI-assisted engineering processes. Training has significant impact on AI adoption efficiency, decision-making

quality, and cost-effectiveness. Without adequate training, organizations risk misuse, over-reliance, or underutilization of AI tools, leading to inefficiencies or defective engineering outputs. Conversely, well-trained personnel can strategically leverage AI to enhance productivity, innovation, and risk mitigation across systems engineering processes.

4. Project Cost Modeling of AI Assistance Usage

Our cost modeling research aims to quantify the impact of using generative AI in systems and software engineering processes [5]. It addresses new challenges in systems and software cost modeling with model definitions, a lifecycle standard, and data analysis process for calibration. This includes new cost factors and size measures to incorporate into existing parametric cost models, requiring empirical data collection and analysis for model calibration. We are gathering data on generated solution sizes, actual effort, and effort without AI assistance for standard systems engineering activities.

To address the cost impacts we have developed a road map for advancing the cost models by leveraging existing modeling and measurement frameworks. We are using the COCOMO cost model framework and calibration procedures [8]. We are also using the comprehensive ISO/IEC/IEEE 15288 lifecycle standard for systems and software engineering [6] for data collection and analysis of detailed activities.

The cost modeling and measurement framework incorporates a new factor for *AI Usage* with a defined rating scale, and data analysis process to calibrate it. An online data collection and Delphi survey to improve the model with expert judgment has also been developed for the community.

The initial rating scale for the cost driver *AI Usage* at the project level shown in Table 2 has been defined using the COCOMO and COSYSMO framework. It consists of five ratings from Very Low to Very High corresponding to the degree of *AI Usage* on a project. These settings are consistent with existing cost factors in COCOMO and COSYSMO. The default setting is Nominal corresponding to a typical project. Data collection is underway to calibrate effort multipliers for each rating level. An example using its effort multipliers is shown in the investment analysis in Section 6.

Very Low	Low	Nominal	High	Very High
Minimal to no AI assistance.	Moderate AI usage, typically for summarization, basic information retrieval, and model templating.	Regular use of AI tools for various tasks like MBSE help, design insights, or testing assistance.	Frequent and strategic use of AI assistance.	AI tools are deeply ingrained in most activities. They are crucial for decision making, problem solving, and automating tasks.
Development relies on traditional methods and tools.	AI tools are not deeply integrated into the development workflow.	AI tools are a recognized part of the toolkit but aren't central to development	AI tools play a central role across the lifecycle from architecting, to design, implementation, and V&V.	The development process is designed around maximizing AI tool benefits.
AI tools may be present but are rarely, if ever, consulted.				

Table 2. *AI Usage Cost Factor Rating Scale*

The project ratings align with the maturity level practices per Table 2 that need to be in place to support the AI adoption. However, each project may utilize the organizational assets in varying degrees so that its project rating could be less than the organizational maturity level. It is also possible that a project can develop its own more advanced or customized infrastructure. These model descriptions may evolve so that the practices in Table 1 become an aspect of *AI Usage* rating in Table 2. The project level ratings may be considered as a weighted average of the underlying AI maturity practices being used on a project.

Already there is very strong convincing data that substantial labor can be saved in steady-state AI tool usage by individuals and teams. Initial empirical data indicates that organizations can achieve substantial cost savings by aligning their development processes with AI capabilities [5].

5. Case Studies

Empirical case studies of systems and software engineering processes are the underpinnings for insights into the cost implications of AI assistance. A variety of data sources are being drawn from to support the iterative model development and calibration. These studies are crucial in validating the cost models and refining parameter calibrations, ensuring they accurately reflect real-world impacts. Different mixed methods for data collection and analysis supporting the model development include:

- Multi-project data collection in conjunction with other cost factors which was used to develop COSYSMO [7] and COCOMO II [8]
- Controlled group experiments performed for this research and from public sources
- Small-scale empirical case studies in the classroom and performed by researchers
- Delphi surveys for expert judgment
- Bayesian approaches combining empirical project data and Delphi results

The initial case studies performed by the authors and students in classroom projects collected data on the actual effort and estimated effort to develop small to medium size software programs and SysML models [4][5]. These indicated effort gains from using AI assistance ranging from about 2 to 10. Though useful approximations, more detailed and rigorous data collection is necessary to replace estimated effort with actual measurements across groups of system engineers using AI assistance or not.

One significant public study involved controlled experiments utilizing GitHub Copilot collecting effort with and without AI assistance [10]. Two large groups of experienced software developers were tasked with the same problem to write a JavaScript web server. One group was assisted by Copilot and the other relying solely on traditional methods. The results demonstrated that developers using Copilot achieved an average reduction of 55% in coding effort while also exhibiting a higher probability of producing correct solutions.

Funded research with the Naval Research Program is underway to updating the COSYSMO model for AI assistance with supporting experiments and data collection with the Naval Information Warfare Center (NIWC). Controlled experiments with systems engineers performing standard tasks per the 15288 standard [5] are being conducted with and without AI assistance, and results measured for effort, time and quality. Current experiments involve requirements analysis and verification, and interpretation of SysML architectural models. Subsequent experiments will involve the generation of SysML v.2 models.

Beyond controlled experiments, multi-project data collection is being employed with additional cost factors across a broad set of real-world projects. COCOMO II and COSYSMO were similarly derived from databases for large team projects with actual effort, schedule and all cost driver ratings. Multi-project data collection in conjunction with other cost factors is going forward for the COCOMO III model [4]. The Boehm Center for Systems and Software Engineering is administering the COCOMO III research and including the *AI Usage* cost factor.

Similar to how COSYSMO and COCOMO II were iteratively developed, the updated models are being calibrated with historical project data as well as expert judgment data via Delphi surveys [7][8][11]. We are combining the project data with the expert judgment data, using Bayesian statistical techniques that balance expert data and historical data [11]. The method leverages prior knowledge to refine the predictive models, and often involves multiple rounds of a Delphi survey as empirical data collection progresses.

Delphi surveys capture what seasoned experts believe are the influences on project effort. They can be used to fill in data gaps in the empirical data. A preliminary Delphi was performed internally to derive the provisional effort multipliers used in the Section 6 investment example.

We have developed an online Delphi survey form to capture both expert judgment and actuals data to help calibrate the model. It is available at <http://softwarecost.org/data/ai> [9]. The data collection is being supported by the Boehm Center for Systems and Software Engineering. We are early in the initiative and the community is highly encouraged to provide feedback on the model definitions, submit data and feedback on the data collection.

6. Example Investment Analysis

This example assesses an organizational investment in systems engineering processes and infrastructure to improve its generative AI capabilities. It is desired to estimate the annual cost savings after adoption and break-even point when the investment will pay for itself. Systems engineering effort in COSYSMO is estimated per Equation 1. This back-of-the-envelope analysis uses the baseline COSYSMO model with provisional calibration values for the *AI Usage* factor in Figure 1 from a preliminary Delphi survey.

$$Effort = A * Size^B * \prod_{i=1}^N EM_i \quad (1)$$

Where

- *Effort* is in Person-Months (PM)
- *A* is a constant derived from historical project data
- *Size* is a sum of weighted system requirements, interfaces, algorithms and scenarios
- *B* is an exponent for the diseconomy of scale
- *EM_i* is an effort multiplier for the *i*th cost driver. The geometric product of *N* multipliers is an overall Effort Adjustment Factor (EAF) to the nominal effort.

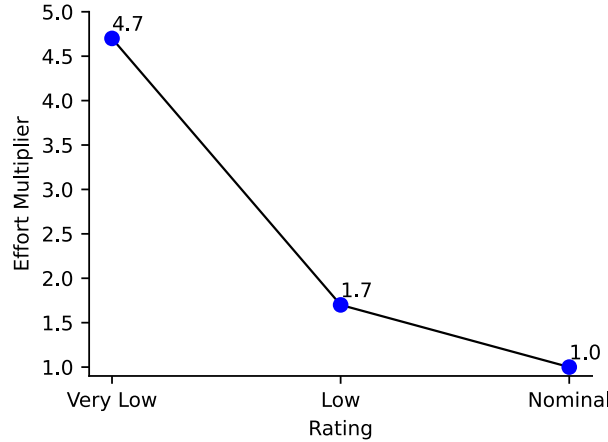


Figure 1. Provisional AI Usage Effort Multipliers for Investment Analysis.

A systems engineering organization of 100 people is currently at maturity Level 2 (Developing) and wants to advance to Level 3 (Competent). They develop 1500 Equivalent Nominal System Requirements per year on projects at \$20,000/Person-Month. Their project practices will improve from a Low rating of *AI Usage* to Nominal. The costs to achieve Level 3 are:

- LLM team will retrain organizational models: 3 people * 3 months * \$20K/person-month = \$180K
- High performance computers purchased for LLM training: \$150K
- All systems engineers take generative AI short course training for three days: 100 people * 3 days * \$20K/person-month = \$300K.

The current value of the *A* constant in COSYSMO without *AI Usage* is .254. When including *AI Usage* and its provisional multipliers, the constant becomes *A* = .054 reflecting the nominal usage of AI. It assumes that setting *AI Usage* to Very Low is the equivalent to cost factors remaining Nominal for the current COSYSMO without *AI Usage*, thus *A* = .254/4.7 = .054. Per the model definition, the *AI Usage* effort multiplier for Nominal is 1.0 and the provisional effort multiplier for Low is 1.7. The *B* exponent in COSYSMO is 1.06. Assuming that all other cost factors are rated Nominal for simplicity, the estimated annual costs for the current and future are calculated with the following.

As-Is (AI Usage = Low)

$$Effort = A * Size^B * EAF = .054 * 1500^{1.06} * 1.7 = 213.6 \text{ PM}$$

$$\text{Annual Cost} = \$20,000/\text{PM} * 213.6 \text{ PM} = \$4.27\text{M}$$

To-Be (AI Usage = Nominal)

$$Effort = A * Size^B * EAF = .054 * 1500^{1.06} * 1.0 = 125.6 \text{ PM}$$

$$\text{Annual Cost} = \$20,000/\text{PM} * 125.6 \text{ PM} = \$2.51\text{M}$$

From the above, the annual savings and break even point can be calculated:

$$\text{Annual Savings} = \$4.27\text{M} - \$2.51\text{M} = \$1.76\text{M}$$

$$\text{Break Even Point} = (\$180\text{K} + \$150\text{K} + \$300\text{K}) / \$1.76\text{M} = \$630\text{K} / \$1.76\text{M} = .36 \text{ years}$$

Hence, the investment of \$.63M to increase by one level will pay for itself in slightly more than four months after adoption on systems engineering projects. This would be a worthwhile investment and the ROI would be a substantial multiplier after a few years.

7. Conclusions and Future Work

Generative AI is already transforming systems engineering by enabling greater efficiency, creativity, and decision support. The integrated maturity and cost modeling framework offers a structured pathway for organizations seeking to adopt generative AI in their engineering processes. By progressing through the maturity levels and using the *AI Usage* cost factor to assess financial impacts, organizations can enhance their engineering capabilities, optimize costs, and remain competitive in an increasingly complex technological landscape.

The integrated maturity and cost modeling framework provides several benefits to organizations adopting generative AI in systems engineering:

- It serves as a roadmap for AI adoption, helping organizations understand where they currently stand and what steps are needed to advance.
- The framework provides a basis for benchmarking progress and identifying best practices, fostering a culture of continuous improvement.
- The integration of a cost model enables better informed cost and schedule estimates for projects using generative AI.
- It supports holistic decision-making that considers both technical capabilities and financial sustainability.

We are early in the cost modeling initiative and the community is highly encouraged to provide feedback on the model definitions, submit data and feedback on the data collection. Community support is imperative to develop the new models. We will eventually provide open source cost modeling tools with new factor(s) in the models for public usage.

Further analysis of AI tool impacts will be performed across the lifecycle aligning artifacts and effort data with ISO/IEC/IEEE 15288 systems and software engineering phases and activities [6]. This lifecycle standard harmonization will also help address large scale team and enterprise processes assisted with AI.

The parametric cost models will be continually refined for AI-enabled productivity enhancements as processes evolve. Ongoing empirical calibrations will ensure that the cost models remain relevant as AI tools become increasingly integrated into systems and software engineering workflows.

Future research could also focus on developing specific assessment tools to help organizations determine their current maturity level and identify targeted actions for advancement. Additionally, detailed case studies of organizations that have successfully integrated generative AI could provide valuable insights into best practices and common challenges. Further exploration is needed to refine the cost model

and calibrate the *AI Usage* cost factor based on empirical data from different systems engineering contexts, as well as to evaluate the long-term financial and operational impacts of AI adoption.

Acknowledgements

This work has been supported by the Naval Research Program at the Naval Postgraduate School, the Naval Information Warfare Center (NIWC), and the Boehm Center for Systems and Software Engineering. The authors also thank Dr. John Wood at NIWC for his support and assistance in the research.

References

1. Paulk, M. C., Curtis, B., Chrissis, M. B., & Weber, C. V. Capability Maturity Model for Software, Version 1.1. Software Engineering Institute, Carnegie Mellon University; 1993
2. CMMI Product Team. CMMI for Development, Version 1.3. Software Engineering Institute, Carnegie Mellon University; 2010.
3. Software Engineering Institute. Cybersecurity Maturity Model Certification (CMMC) 2.0. Software Engineering Institute, Carnegie Mellon University; 2021.
4. Madachy, R., Bell, R., Longshore, R. "Factoring AI Assistance into Software and Systems Engineering Cost Models", Proceedings of the 2023 International Boehm COCOMO Forum; 2023.
5. Madachy, R., Bell, R., Longshore, R. "Systems Acquisition Cost Modeling Initiative for AI Assistance." Proceedings of the 2024 Acquisition Research Symposium, Monterey, CA.
6. International Organization for Standardization. ISO/IEC/IEEE 15288:2015 Systems engineering – System life cycle processes. International Organization for Standardization; 2015.
7. Valerdi, R. "The Constructive Systems Engineering Cost Model (COSYSMO)." PhD thesis, University of Southern California; 2005.
8. Boehm, B., Abts, C., Brown, W., Chulani, S., Clark, B., Horowitz, E., Madachy, R., Reifer, D., & Steece, B. *Software Cost Estimation with COCOMO II*. Prentice-Hall; 2000.
9. softwarecost.org. AI Assistance Usage Delphi Form, <http://softwarecost.org/data/ai/>, 2024.
10. GitHub. "Research: quantifying GitHub Copilot's impact on developer productivity and happiness". <https://github.blog/2022-09-07-researchquantifying-github-copilots-impact-on-developerproductivity-and-happiness>, 2022.
11. Chulani, S., Boehm, B., & Steece, B. "Bayesian Analysis of Empirical Software Engineering Cost Models." *IEEE Transactions on Software Engineering*, 1999;25:4, pp. 573-583